

Student postures and gesture recognition system for adaptive learning improvement

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Abstract— Adaptive learning is an educational method that uses computers to coordinate interaction with student and provide customized learning resources and activities to meet each student's unique needs that are a trend. In order to adjust the content and teaching methods appropriate for each learner, it is necessary to tract the attitude of student with the content and methods being implemented. One of the ways of detecting student's attitudes is through the gestures and posture of student in the classroom.

In this study, we propose to develop a model to monitor and identify students' postures and gestures during class hours to help assess the level of student participation in the content and teaching methods of teacher. From there, it is possible to give advice to trainers and training managers in adjusting the subject content, teaching methods suitable for each specific object.

Keywords— *Posture recognition, Gesture recognition, emotion recognition, adaptive learning*

I. INTRODUCTION

In recent years, detecting student emotion and actions in the classroom is a key topic for researchers who focus on intelligent training systems. Together with the development of advanced techniques and the power of computing ability, many solutions for these problems are presented and gained some positive results. Knowing exactly whether a student is interested to the lesson or not in the classroom is very important for designing adaptive learning system and intelligent tutoring systems. Besides this, detecting student engagement supports teachers (in a traditional classroom) or tutoring systems in tracking learner attention. The success of a traditional course or online learning course depends on the outcomes of students, and that results are related to the engagement of the learners. Some researches show that engagement detection makes opportunities for improving learning and teaching [1][2] and even for issuing targeted interventions of the monitors or educational manager [3].

In the firm of engagement detection, according to Dewan [4], he divided engagement detection into their categories, such as Automatic, Semi-Automatic and Manual. Toward computer vision-based point of view, he mentioned that Facial Expression, Gesture, and posture, Eye movement are three facets in automatic detection.

Facial expression recognition receives many concerns from researchers. Many researches concentrate on student's face to

guess the emotions of learners. Nezami et al. [5] used a convolutional neural network to classify emotion of students, based on the pre-trained result from FER-2013 dataset [6]. Whitehill et al. [3] divided student engagement into four levels from 1 to 4 and used support vector machine (SVM) classifier to detect engagement. Bosch in [7] focused on mind wandering a type of cognitive disengagement. He concentrated on extracting low-level and high-level features from video clips for the face recognition process.

Eye movement is interested in recent years, Raina [8] proposed an eye-tracking-based model which used eye-tracker equipment to compare engagement between two student groups to reduce content skipping when they study on the online environment. Krithika and Lakshmi [9] built a system call SERS (Student Emotion Recognition System) that can identify and monitor emotions of the student in an e-learning environment and provide a real-time feedback mechanism to enhance the e-learning aids for better content delivery. The system can detect eyes, head movement for estimating student engagement level.

Our work focuses on the third branch: Gesture and postures because this kind of non-verbal communication can express various student states in the classroom. Gesture and postures are two important forms of non-verbal communication through our body language [4].

In this paper, we deal with engagement during learning via technology. In this model, the students in a classroom are supposed to sit through a lesson, thus student gestures constitute a motion space consisting of the upper body, face, and hands. The system is setup with one camera in front of the classroom, or two cameras in left and right corner in front of the classroom. We also build a dataset for classifying student gesture and posture in the classroom to identify whether a student is interested or not in learning lesson.

We use a convolutional neural network to train my new data. Before this, the network was pre-trained to get the standard weighted matrix for ordinary actions.

The paper is organized as follows. After the introduction and related work section, the propose model is presented in section III. Then is the experiment and result in Section IV. Finally, the paper is concluded in Section V.

II. RELATED WORK

Many computer-based approaches have been used to detect engagements with gesture and posture information. Mitra and Acharya [9] divided the gestures arising from different body parts into three classes: hand and arm gestures, head and face gestures, and body gestures. Wan and Sawada [10] used the probabilistic distribution of the arm trajectory to classify 30 different hand gestures. This results show that the recognition system can obtain high accuracy of “Rain” and “Round” gestures. However, nine markers should be attached to indicate the feature points of a student’s body. This assumption is difficult to implement in a classroom. Fang et al. [11] classify six popular gesture of student in classroom which are “raising the left hand”, “raising the right hand”, “raising two hands”, “standing up”, “lying prone”, and “normal posture” to help the teacher notice some behaviors of the students in teaching time. Grafsgaard et al. [12] measured the postural quantity of motion of hands and head to find the relationship between nonverbal behaviors and dialogue and its effects while learning. Potnis and Jahagirdar [13] captured videos and detected hand gesture in the classroom to convey it to the lecturer. Sathyanarayana et al. [14] built a dataset called SDMATH and used HOG and SVM to localize deictic gestures in a teaching learning session. Zaletelj and Košir [15] measured students’ attention level in classroom by using Kinect device. They combined upper body posture, face gaze points, and facial features and used seven classifiers for classification, then compared results between these methods. Rich et al. [16] used gesture and speech as one channel to follow up the user engagement during human-robot interactions. Klein and Celik [17] built a system called WITS (Wits Intelligent Teaching System) to support lecturers to track student’s engagement by monitoring their real-time feedback. They created an Interesting Map which pointed out students who are engaged or disengaged in the classroom in the lecture time.

III. PROPOSAL MODEL

This paper propose a student gesture and posture recognition model employed in a classroom with one is setup at the front of the classroom or two cameras is setup at the left and right corner in front of the classroom to capture video sequences. Figure 2 show example of picture from one camera is set in front of the classroom, while figure 3, 4 are example of image capture from right and left corner at the front of classroom.

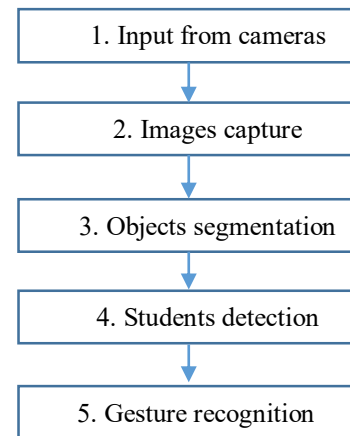


Fig. 1. Student gesture recognition model

Students are assumed to sit stationary in a position for the duration of a lesson, thus student gestures constitute a motion space consisting of the upper body, face, and hands.



Fig. 2. Example of an input sequence from front-camera



Fig. 3. Example of an input sequence from right-camera



Fig. 4. Example of an input sequence from left-camera

Firstly, input sequences from camera are extracted for a certain period (example 2 images/minutes). We do not use real-time sequence because the student's gesture is stable for a certain during. Secondly, images are extracted from camera are take into object detection system. In this paper, we propose using YOLOv3 [18] to extract object. Each object is bounded in a box. Then, we extract small image from origin image for each object such that the type is person. Each small picture of a student then is put into gesture recognition. The figure 5 illustrates object detection from YOLOv3. Yellow box is student, while orange box is chair.



Fig. 5. Example of object detection from a frame

The heart of the student gesture recognition model is Gesture recognition system. In this paper we propose using pre-train model and

IV. EXPEIMENT AND RESULT

A. Dataset

Related to this paper, we built a dataset from several classes in the lecture time. This database is built by extracting images from high-density videos which were captured by some fixed angle cameras in the classroom. These cameras were secret to the student to ensure capturing the natural gesture and posture of students. According to popular postures of a student in the

lecture time, several posture and gesture can be considered as main actions and they were labeled as interest (engagement) and non-interest (disengagement). Data after collecting was handily processed such as cropping, reducing resolution and then labeling manually.

The dataset includes 942 files which are divided into 8 classes in 2 groups. The first group is labeled as interesting (engagement) and non-interest (disengagement) for the others. With the interest group, we continuously divided into some actions such as "Writing", "Reading", "Raising hand", "Concentrating". On the other hand, with the non-interesting group, we split into "Using telephone", "Looking up", "Looking around", "Head down".

The labeled data was divided into two sets, training set and test set with the rate of 80% and 20%, respectively



Fig. 6. List of student guesture in classroom

B. Experiment

In recent years, deep learning shows some advantages compare to other learning methods and artificial intelligence. Especially, Convolutional neural network is one of the deep structure with archive state-of-the-art result in various domains. The main components of this network include some convolutional layers, Max-pooling layers, Fully-connected layers and a soft-max layer to identify the number of classes. In this paper, we construct a convolutional neural network with the input is a series of 224x224 images which are resized from the origin.

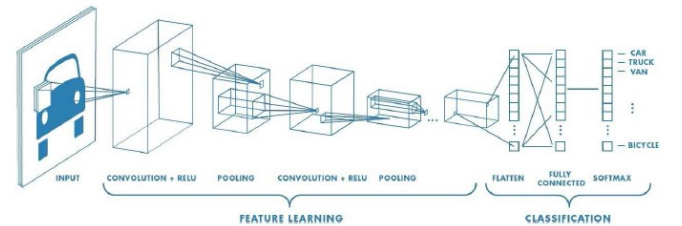


Fig. 7. A CNN structure sample from Toward Data Science[17]. Accessed on June, 2019

First, we re-use deep convolutional network based on 2 structures VGG16 and VGG19 [19]. The current network is also pre-trained on the ImageNet [20] dataset which contains 1.2 million high-resolution images [21] for speeding up convergence. Because of the small dataset, we apply transfer learning [22] and use data augmentation techniques as in [23] to improve overall accuracy and avoid overfitting. Some augmentation techniques like re-scaling, flipping, cropping, rotating and shifting are applied. Learning rate is set to 0.0001 and momentum of the network is 0.9 (based on default values of Keras library).

Secondly, the first trained model is pop out by freezing some first layers (train only last 5 layers) for the next classification process.

Next steps, we use regularization technique with dropout rate is 0.2.

Finally, 8 classes which are equivalent to 8 gesture and posture type in the classroom are classified by a soft-max activation function. Our proposed procedure was mentioned as below:

Step 1: Create a new model which inherited the architecture of VGG.

Step 2: Freeze some first layers of the new model.

Step 3: Use weights matrix from ImageNet.

Step 4: Use data augmentation with new dataset.

Step 5: Re-compile and continue to train pre-trained model with small target dataset.

Our experiment is conducted on an Intel Xeon based system with an 8Gb memory GPU card GeForce GTX 1070 and CUDA 9.0 library. Keras framework with TensorFlow backend is chosen for developing our convolutional network. The structure of this network is represented as Fig.8.

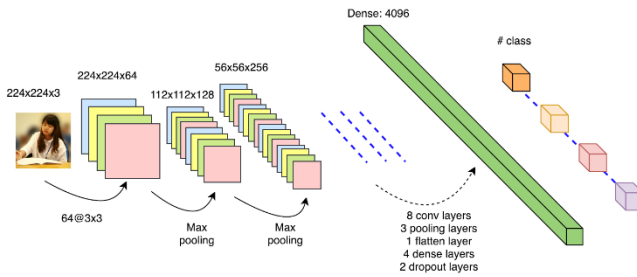


Fig. 8. Convolutional neural network architecture. First layers are based on VGG16 and VGG19 structures. Five top layers are retrained with new parameters.

C. Result

Although the comparison between researches is not appropriate because of the difference of the datasets and methods, we just show the accuracy of the other similar papers to prove that our achievement is acceptable when applying very lightweight dataset. In our experiment, our accuracy gained

80.8% compared to [11] gained 72%. Klein [17] showed impressive result with the accuracy was 89.7% after 20.000 training epochs and over 200.000 images.

TABLE I. COMPARE RESULT ON SOME APPROACHES

Method	Accuracy
Training CNN from scratch	68.8%
Training CNN with VGG16	74.9%
Training CNN with VGG19	76.0%
Training CNN with VGG19 and applied transfer learning	80.8%

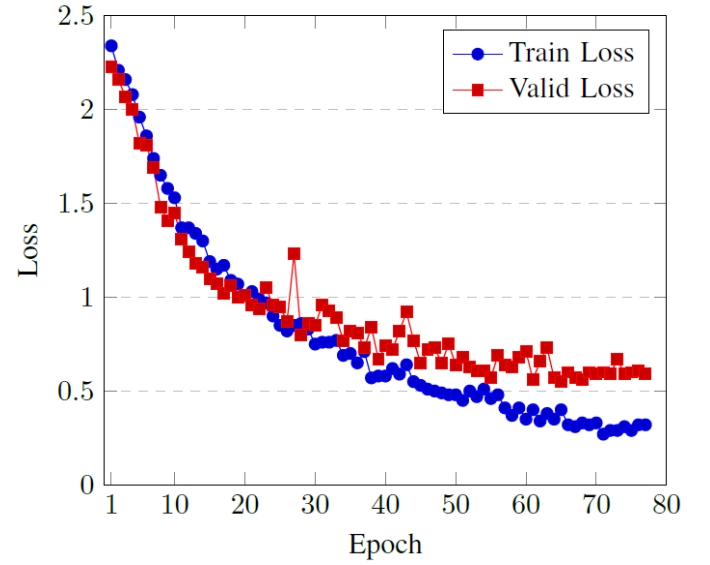


Fig. 9. Loss curves of training and validating process

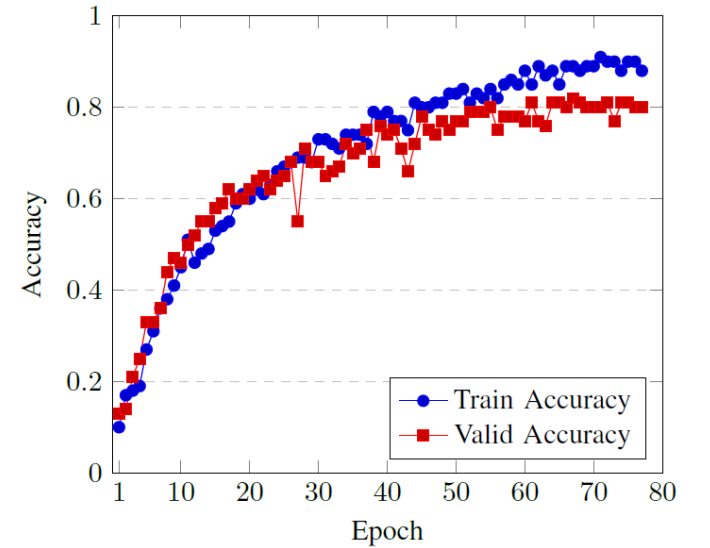


Fig. 10. Accuracy curves of training and validating process

Fig 9 and 10 show loss curves and accuracy curves, respectively. After approximate 80 iterations, the validating loss/accuracy does not decrease/increase. The early stopping trigger activate the immediate stopping process to prevent model overfitting. One of the reasons why the result is not high enough compared to [17] because of the ambiguity of gesture and posture of the student.

V. CONCLUSION

Posture and gesture are fundamental components of embodied affect which are tightly related to student's state in the classroom. Toward the next generation of intelligent training systems, the ability of analysis correctly student's attitude may help to support both teachers and manager in improvement quality of training services.

In this paper, we have proposed a model for student's posture and gesture recognition system. It can be applied in an adaptive learning system to improve quality of training services. The result also shows that applying transfer learning and convolutional neural network can classify effectively student gesture and posture in the classroom based on predefined actions labels. In the future, to improve the overall accuracy of the classification process by expanding the dataset more samples. Besides this, we can build more convolutional neural networks to form multi-streams CNN for incorporating complementary features of motion networks.

Last but not least, the results of this model is only the first step of an intelligent training system. To support teachers in the classroom or some of the e-learning environments, we need to build more components such as domain knowledge model, student trait model, human-machine interface and other necessary modules for a complete intelligent training system.

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